

# Week 3 - Day 4

## Classification Model Evaluation

5-Minute Review Notes + Practical Exercise Summary

### Day Goal

Evaluate classification models beyond accuracy using Precision, Recall, F1 Score, classification reports, and decision thresholds. The key idea is that the best metric depends on which mistakes matter for the real problem.

**Day 4 Status:** COMPLETE

### 1. Why Accuracy Is Not Enough

Accuracy measures the fraction of all predictions that are correct. It is useful, but it can hide serious failures when classes are imbalanced.

Accuracy = Correct Predictions / Total Predictions

990 legitimate + 10 fraud  
Predict everything as legitimate -> 990/1000 = 99% accuracy  
Fraud detected = 0/10

A model can have high accuracy while failing at the task we actually care about.

### 2. Precision

**Precision:** Out of everything predicted as positive, how many were actually positive?

Precision =  $TP / (TP + FP)$

High Precision means positive predictions are trustworthy. It is especially important when False Positives are costly.

### 3. Recall

**Recall:** Out of all actual positive cases, how many did the model successfully find?

Recall =  $TP / (TP + FN)$

High Recall means fewer real positives are missed. It is especially important when False Negatives are costly.

### 4. Precision vs Recall

| Metric    | Starts From         | Question                                       |
|-----------|---------------------|------------------------------------------------|
| Precision | Predicted positives | When I predict positive, how often am I right? |
| Recall    | Actual positives    | Of all real positives, how many did I find?    |
| F1        | Precision + Recall  | How well are both balanced?                    |
| Accuracy  | All predictions     | How many were correct overall?                 |

## 5. F1 Score

F1 combines Precision and Recall using the harmonic mean. It is useful when both metrics matter and we want one balanced score.

```
F1 = 2 * (Precision * Recall) / (Precision + Recall)
```

## 6. Classification Report

```
from sklearn.metrics import classification_report
print(classification_report(y_test, y_pred))
```

The report shows Precision, Recall, F1 Score, and Support for each class. **Support** is the number of actual examples in that class; it is a count, not a score.

## 7. Day 4 Practical Model

The exercise predicted whether a student would Pass (1) or Fail (0) using Study Hours, Attendance, and Assignments Completed.

```
X = df[["Study_Hours", "Attendance", "Assignments_Completed"]]
y = df["Pass"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

model = LogisticRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

**Target:** Pass = 24/40 = 60%; Fail = 16/40 = 40%. Somewhat imbalanced, but not severely imbalanced.

## 8. Confusion Matrix

```
[[4, 0],
 [1, 5]]
```

| Item | Count | Meaning                     |
|------|-------|-----------------------------|
| TN   | 4     | Actual Fail, Predicted Fail |
| FP   | 0     | Actual Fail, Predicted Pass |
| FN   | 1     | Actual Pass, Predicted Fail |
| TP   | 5     | Actual Pass, Predicted Pass |

The model made one mistake: one student who actually passed was predicted as Fail.

## 9. Evaluation Results

| Metric    | Result | Interpretation                           |
|-----------|--------|------------------------------------------|
| Accuracy  | 0.9000 | 9 of 10 test predictions were correct.   |
| Precision | 1.0000 | Every predicted Pass was actually Pass.  |
| Recall    | 0.8333 | Found 5 of 6 actual passing students.    |
| F1        | 0.9091 | Precision and Recall are strong overall. |

## 10. Classification Thresholds

Logistic Regression outputs probabilities. A threshold converts Class 1 probability into a final class prediction.

```
y_prob = model.predict_proba(X_test)[:, 1]
y_pred_03 = (y_prob >= 0.3).astype(int)
y_pred_05 = (y_prob >= 0.5).astype(int)
y_pred_07 = (y_prob >= 0.7).astype(int)
```

| Change          | Typical Effect                                                                 |
|-----------------|--------------------------------------------------------------------------------|
| Lower threshold | More positives; Recall may increase; FP may increase; Precision may decrease.  |
| Raise threshold | Fewer positives; FN may increase; Recall may decrease; Precision may increase. |

## 11. Threshold Experiment Result

| Threshold | Precision | Recall | F1     |
|-----------|-----------|--------|--------|
| 0.30      | 1.0000    | 0.8333 | 0.9091 |
| 0.50      | 1.0000    | 0.8333 | 0.9091 |
| 0.70      | 1.0000    | 0.8333 | 0.9091 |

All three thresholds gave the same predictions because none of the model probabilities fell between 0.30 and 0.70. A threshold change affects a row only when its probability crosses the threshold boundary.

Example:  $P(\text{Class 1}) = 0.55 \rightarrow$  threshold 0.50 predicts 1, while threshold 0.70 predicts 0.

## 12. Business Goal Determines the Metric

| Business Concern           | Mistake         | Metric    |
|----------------------------|-----------------|-----------|
| Do not miss real positives | False Negatives | Recall    |
| Avoid false alarms         | False Positives | Precision |
| Need balance               | Both matter     | F1        |
| Overall correctness        | All errors      | Accuracy  |

For the churn question, **Model B** was preferred because Recall = 0.80 versus Model A Recall = 0.45. The goal was specifically to avoid missing customers likely to leave.

## 13. Assessment Corrections

| Q | Correct Review                                                      |
|---|---------------------------------------------------------------------|
| 1 | Precision: of predicted positives, how many were actually positive? |
| 2 | Recall: of actual positives, how many did the model find?           |
| 3 | TP=40, FP=10, FN=20 $\rightarrow$ Precision=0.80; Recall=0.6667.    |
| 4 | Low False Negatives $\rightarrow$ Recall is being prioritized.      |
| 5 | Legitimate email predicted Spam = False Positive; watch Precision.  |
| 6 | F1 combines Precision and Recall.                                   |

| Q  | Correct Review                                                                         |
|----|----------------------------------------------------------------------------------------|
| 7  | Support = actual count in each class.                                                  |
| 8  | Lower threshold -> more positives; Recall may rise; Precision may fall.                |
| 9  | Higher threshold can turn actual positives into negatives -> FN rises -> Recall falls. |
| 10 | Model B because higher Recall catches more actual churners.                            |

## 14. Quick Review Cheat Sheet

Accuracy -> overall correctness  
Precision -> trust positive predictions  
Recall -> find actual positives  
F1 -> balance Precision + Recall

$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$   
 $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$

Lower threshold -> more positive predictions -> Recall may rise  
Higher threshold -> fewer positive predictions -> Recall may fall

FP matters -> think Precision  
FN matters -> think Recall

## 15. Completion Summary

You built and evaluated a Pass/Fail Logistic Regression model, interpreted its confusion matrix, calculated Accuracy, Precision, Recall and F1, read a classification report, and experimented with decision thresholds.

The most important lesson: **the best model metric depends on the real-world cost of False Positives and False Negatives.**

**Week 3 Progress:** 4/7 complete

**Completed:** Day 4 - Classification Model Evaluation

**Next:** Week 3 - Day 5